CSE 350

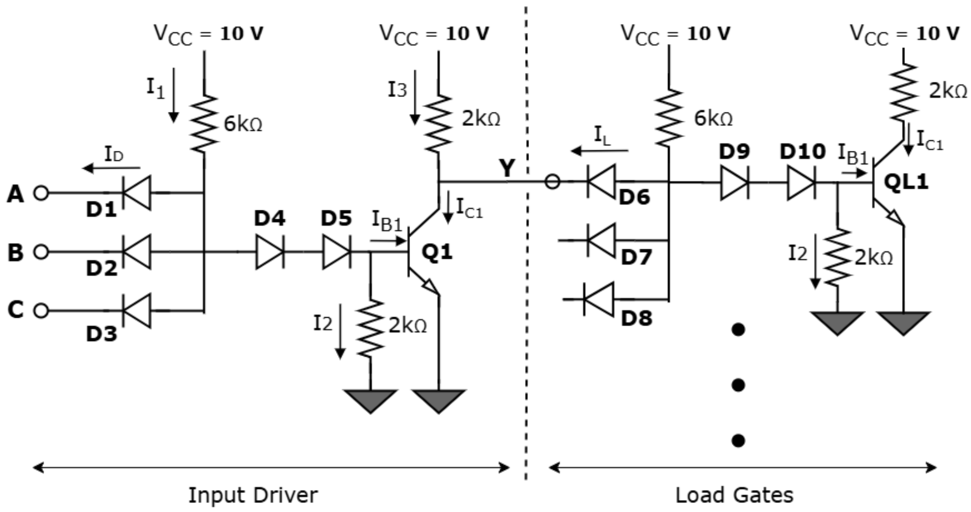
Digital Electronics and Pulse Techniques

Assignment-03 — Total – 20 marks

Question 1 — 05

Implement the logic function, $Y = \overline{A+B}\,.\overline{C}$ using **any combination of the logic families** you know. Then, mention the **operating mode** of your diodes/transistors when the **output is high**. [*You may simplify the function if possible. Mentioning the values of circuit components is not necessary*.]	[5]

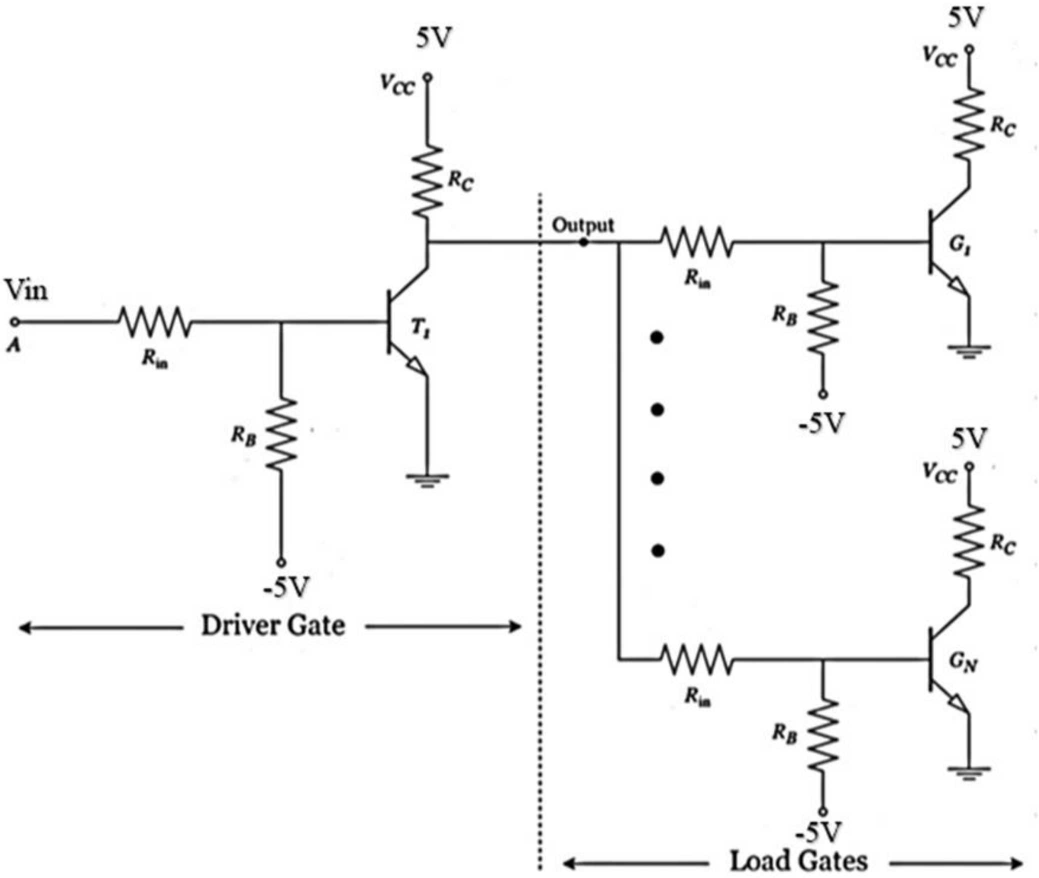
Question 2 — 05



For the above circuit, $B_F = 30$, V_{CE} (sat) = 0.1 V, V_{BE} (sat) = 0.8 V (a) Find $\mathbf{I_1, I_2, I_{B1}, I_{C1}}$ and $\mathbf{I_D}$ if any input is logical low (0.1 V) for the driver circuit when **load gates are not connected**. Then also find out the power dissipation (of the driver) for this case. (b) Find $\mathbf{I_1, I_2, I_{B1}, I_{C1}}$ and $\mathbf{I_D}$ if all input is logical high (10 V) for the driver circuit when **load gates are not connected**. Then also find out the power dissipation (of the driver) for this case. Prove that Q1 transistor is in Saturation mode. (c) The driver circuit is now connected with Load gates. Calculate maximum fanout of the overall DTL setup.	[1+2+2]

Question 3 **05**

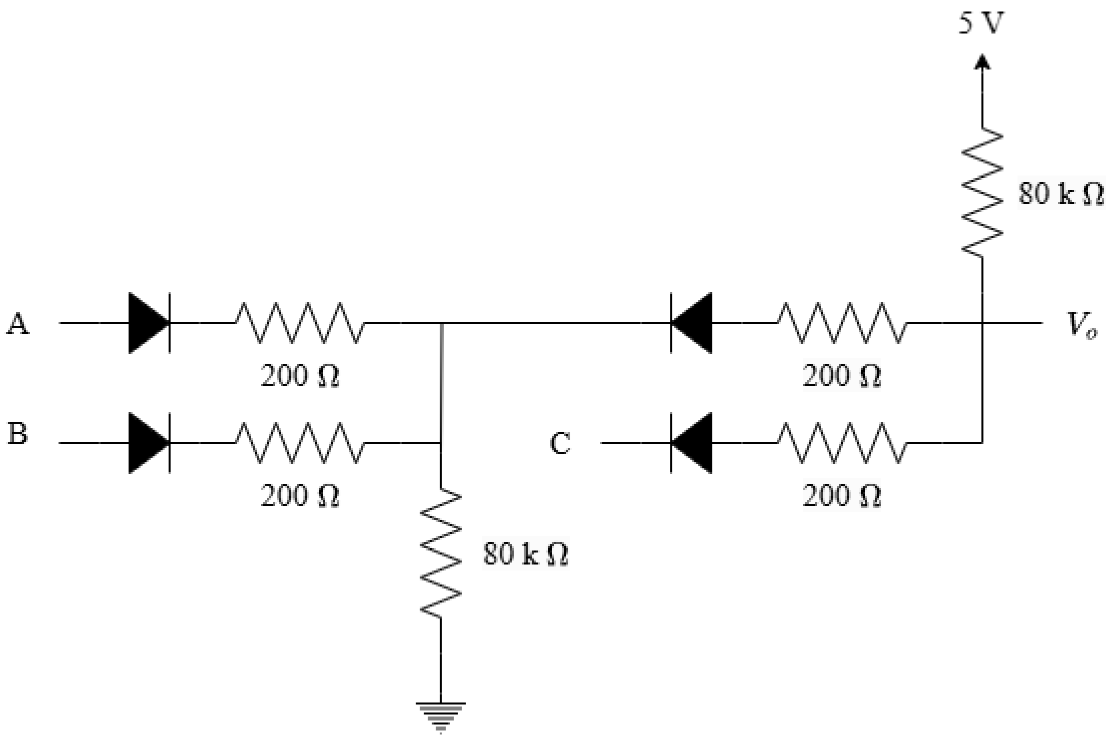
For the given circuit assume $V_{OH} = 4.5V$ and $V_{OL} = 0.2V$. Also assume common emitter current gain, $\beta_F = 30$. $V_{BE}(sat) = 0.8$ V, $V_{CE}(sat) = 0.2$ V and cut in voltage for transistor $V\gamma = 0.5$ V.



Driver and Loads are identical where **R_{in} = 5k**, **R_B = 100k**, and **R_C=2k.**

(a)	If **Vin = High**, find the power dissipation in the **Driver** circuit assuming **48 loads** are connected.	2.5
(b)	Find the value of **Noise margin** of the circuit.	2.5

Question 4



For the circuit shown, input voltage,

$V_{in}(high) = 5\ V, V_{in}(low) = 0\ V$

(c)	CO2	What logic function does the above circuit implement?	[2.5]
(b)	CO2	Find the output voltage & power dissipation of the entire circuit when A=low, B=high, C=low	[2.5]